

HDOC Day-28 Assignment Solutions

Course: Computer Programming (C Language)

Topic: Variable-Length 2D Arrays and Array Layout Manipulation

Question 1

Write a C program to:

- Read the number of rows and columns from the user. Ensure that both values are positive. If not, repeat to get the desired input(s).
- Declare a variable-length 2D integer array using the given dimensions.
- Read the array elements column by column using nested loops, with the outer loop iterating over columns and the inner loop over rows.
- Print the elements in row-major order using a single loop and a linear index.

```
#include <stdio.h>

int main() {
    int rows, cols;

    // a) Read rows and columns with positive validation
    do {
        printf("Enter positive number of rows: ");
        if (scanf("%d", &rows) != 1) {
            while (getchar() != '\n'); // clear buffer
            rows = 0;
        }
    } while (rows <= 0);

    do {
        printf("Enter positive number of columns: ");
        if (scanf("%d", &cols) != 1) {
            while (getchar() != '\n'); // clear buffer
            cols = 0;
        }
    } while (cols <= 0);

    // b) Declare a variable-length 2D integer array
    int arr[rows][cols];

    // c) Read elements column by column (outer loop: columns, inner loop: rows)
    printf("Enter elements column by column:\n");
    for (int j = 0; j < cols; j++) {
        for (int i = 0; i < rows; i++) {
            printf("Element at [%d][%d]: ", i, j);
            scanf("%d", &arr[i][j]);
        }
    }

    // d) Print elements in row-major order using a single loop and linear index
    printf("\nElements in row-major order:\n");
    int total_elements = rows * cols;
```

```

        for (int k = 0; k < total_elements; k++) {
            int r = k / cols;
            int c = k % cols;
            printf("%d ", arr[r][c]);
        }
        printf("\n");

        return 0;
    }
}

```

Question 2

Write a C program that reads elements into a variable-length 2D array, copies them into a 1D array in row-major order, and displays both arrays.

```

#include <stdio.h>

int main() {
    int rows, cols;

    // Input dimensions
    printf("Enter number of rows: ");
    scanf("%d", &rows);
    printf("Enter number of columns: ");
    scanf("%d", &cols);

    if (rows <= 0 || cols <= 0) {
        printf("Invalid dimensions.\n");
        return 1;
    }

    // Declare variable-length 2D array
    int arr2D[rows][cols];

    // Declare 1D array to hold all elements
    int arr1D[rows * cols];

    // Read elements into 2D array (row-major order)
    printf("Enter elements for the 2D array:\n");
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            printf("Element at [%d][%d]: ", i, j);
            scanf("%d", &arr2D[i][j]);
        }
    }

    // Copy elements into 1D array in row-major order
    int index = 0;
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            arr1D[index] = arr2D[i][j];
            index++;
        }
    }

    // Display the 2D array

```

```
printf("\nDisplaying the 2D array:\n");
for (int i = 0; i < rows; i++) {
    for (int j = 0; j < cols; j++) {
        printf("%d\t", arr2D[i][j]);
    }
    printf("\n");
}

// Display the 1D array
printf("\nDisplaying the 1D array:\n");
for (int i = 0; i < rows * cols; i++) {
    printf("%d ", arr1D[i]);
}
printf("\n");

return 0;
}
```